

# Astroarch Interface

Remote control of an AstroArch observatory from Android



User Manual & Installation Guide

Version 0.2.12

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# 1. Introduction

**Astroarch Interface** is an Android application that lets you fully control an astronomical observatory based on the **AstroArch** distribution (Arch Linux for Raspberry Pi with KStars/Ekos/INDI) running on a Raspberry Pi 5. The app connects to the Raspberry Pi via Tailscale and mirrors all Ekos features in real time using a mobile-friendly UI.

The project is split in two parts:

- **astroarch-bridge backend**: a Python daemon (FastAPI + WebSocket) installed on the Raspberry Pi 5. It talks to the INDI server, PHD2 and Ekos via DBus, exposing a REST API + two WebSocket streams to the app.
- **Android app**: written in Flutter, with 14 dedicated screens covering all the Ekos modules cloned for mobile use.

## 1.1 What you can do

- Real-time view of mount, camera, focuser, filter wheel, dome and weather state (live telemetry over WebSocket)
- Plan and run multi-job capture sequences just like in Ekos
- Run a full pre-flight pipeline: slew → plate solve → sync → guide → autofocus → capture, in one tap
- Search objects by name (SIMBAD) and GoTo with one tap
- Control PHD2 (calibration, guiding, dither) with a live RMS chart
- Control the focuser manually or trigger an iterative autofocus with a live V-curve plot
- Run plate solving via solve-field/astrometry.net and sync the mount on the result
- View captured images live (FITS auto-stretched to JPEG over WebSocket)
- Edit any INDI property from the panel that clones the Ekos INDI Control Panel
- Plan scheduler jobs with twilight/altitude/weather conditions

## 1.2 Architecture

High-level data flow:

```
Android app --Tailscale (WireGuard)--> Raspberry Pi 5
|- astroarch-bridge :8765
|  |- REST /api/*
|  |- WS /ws/state (live clone)
|  |- WS /ws/frames (JPEG live)
|- INDI server :7624
|- PHD2 :4400 (JSON-RPC)
|- KStars/Ekos (DBus)
|- ~/Pictures/Ekos (FITS storage)
```

**Note:** Tailscale provides end-to-end WireGuard encryption. There is no need to set up HTTPS on top: port 8765 is exposed in cleartext but the Tailscale tunnel makes the traffic inaccessible to anyone outside your tailnet.

## 2. Installation

### 2.1 Prerequisites

On the Raspberry Pi (server side):

- AstroArch (Arch Linux for Raspberry Pi) distribution
- Python 3.11+ with pip
- Tailscale installed and active
- KStars/Ekos installed with a configured INDI profile
- PHD2 installed (optional, for guiding)
- astrometry.net + index files in `~/.local/share/kstars/astrometry/` (optional, for plate solving)

On the Android phone (client side):

- Android 8.0 (API 26) or later
- Tailscale app installed and connected to the same tailnet as the Raspberry Pi
- ~50 MB of free space

### 2.2 Backend installation on the Raspberry Pi

#### Step 1 - Transfer the backend

From the computer where you have the *backend/* folder of the Astroarch Interface distribution:

```
scp -r backend/ astronaut@100.74.22.40:/tmp/
```

(Replace 100.74.22.40 with the Tailscale IP of your Raspberry Pi)

#### Step 2 - Install the service

SSH into the Raspberry Pi and run:

```
ssh astronaut@100.74.22.40
cd /tmp/backend
sudo bash deploy/install.sh --user astronaut
```

The installer:

- Creates/uses the specified user (default: *astroarch*; pass `--user astronaut` to use your own)
- Installs the Python dependencies (FastAPI, uvicorn, astropy, Pillow, watchdog, pydantic, websockets, numpy)
- Installs the `astroarch_bridge` package
- Creates the `astroarch-bridge.service` systemd unit
- Enables auto-start on boot
- Starts the daemon

At the end the script prints the URL and the auto-generated **Bearer token**. Save it: you will need it to connect from the app.

```
==> astroarch-bridge installed and running
URL: http://100.74.22.40:8765
Token: kJ3xSn9mZ7TqW... (example)
```

### Step 3 - Verify the service

```
systemctl --user status astroarch-bridge
journalctl --user -u astroarch-bridge -f
curl http://localhost:8765/healthz
```

The healthz output should show:

```
{"ok":true,"version":"0.1.0","indi":"...","phd2":"..."}
```

## 2.3 Desktop dashboard installation

A small Tk dashboard is installed on the AstroArch desktop to monitor the bridge state and generate the QR code for the app:

```
scp -r desktop_dashboard/
astronaut@100.74.22.40:/home/astronaut/astroarch-bridge-dashboard
scp desktop_dashboard/AstroarchBridge.desktop \
astronaut@100.74.22.40:/home/astronaut/Desktop/
ssh astronaut@100.74.22.40 'chmod +x ~/Desktop/AstroarchBridge.desktop'
```

On the AstroArch desktop you will see the **Astroarch Bridge** icon. It opens a window with service status, Ekos info, URL/Token, a QR code for the app, and Connect/Disconnect buttons.

## 2.4 APK installation on Android

On the phone:

- Transfer the AstroarchInterface-v0.2.12.apk file to the phone (USB / Drive / Tailscale Drop)
- Open the file from the file manager. Android will ask you to enable "Install apps from unknown sources" for the file manager: confirm.
- Tap **Install** → the app shows up in the launcher as "**Astroarch Interface**"

## 2.5 First connection

Open the app:

- Tap **SCAN QR FROM DASHBOARD** and frame the QR code shown on the desktop dashboard of the Raspberry Pi: host, port and token will be filled in automatically.
- Or fill them in manually:

|              |                                                                                                 |
|--------------|-------------------------------------------------------------------------------------------------|
| <b>HOST</b>  | Tailscale IP of the Raspberry Pi (e.g. 100.74.22.40)                                            |
| <b>PORT</b>  | 8765                                                                                            |
| <b>TOKEN</b> | The one printed by the installer or read with <code>cat ~/.config/astroarch-bridge/token</code> |

Tap **CONNECT**. If the connection works you go straight to the Dashboard. If it fails, tap **TEST** for step-by-step diagnostics.

**Warning:** If the connection fails with a timeout, make sure Tailscale is active on the phone. Some Android devices (Xiaomi/MIUI, OnePlus, Samsung) need battery optimization disabled for Tailscale, otherwise the VPN gets killed in the background.

## 3. Main screens

The app uses a 5-tab **bottom navigation** always visible (Dashboard, Mount, Align, Capture, Guide) and a **side drawer** with the advanced screens, opened from the menu icon in the AppBar.

### 3.1 Dashboard

The first screen after login. Shows in real time:

- **Connection banner** at the top with INDI / PHD2 / WS state / WS frames (green=ok, yellow=connecting, red=fail)
- **Active target** with RA/Dec coordinates and mount status
- **Last captured image preview** auto-stretched, tap for full-screen Live View
- 4 status cards: Mount (tracking), Camera (temperature), Guide (PHD2 RMS), Focuser (position)
- Observatory telemetry: weather, dome, weather safe

### 3.2 Mount

Full telescope control:

- **Live RA/Dec** updated in real time, pier side, current mount status
- **SIMBAD search**: type a name (M 31, NGC 7000, Vega) and tap SEARCH. RA/Dec resolved via astropy. Buttons: GOTO+TRACK / SLEW / SYNC
- **Manual GoTo** with RA (hours) and Dec (degrees) fields
- **N/S/E/W slew joypad** with rate selection (GUIDE / CENTERING / FIND / MAX depending on the driver)
- Quick Park/Unpark/Sync/Stop
- **Tracking mode** chips: Sidereal / Lunar / Solar / Off

### 3.3 Align (Plate Solve + Polar Align)

**Plate Solve** tab — exact clone of Ekos Align:

- Live FITS preview (pinch zoom) **stretched identically to Ekos**
- Configurable Exposure / Gain / Binning, applied to the camera before the solve
- Solver action chips: GoTo / Sync / Slew to target / Nothing
- Solver mode chips: StellarSolver / Remote (INDI)
- Live telescope coordinates and solution: RA, DEC, Err, PA, Pixel scale, FOV, Focal length, F-ratio
- Solve history with color-coded dRA/dDEC (green<50", yellow<150", red beyond)
- Ekos-style target plot with concentric 50/100/150" rings
- Expandable Ekos Align log

**Polar Align** tab — drift-based 3-step routine:

- Capture + plate solve at 3 different RA positions
- AZ/ALT error computation from Dec drift
- Suggested screw adjustments

## 3.4 Capture

Multi-job sequencer in Ekos style:

- **Cooler panel** with live sensor temperature, editable target, power % bar, visual ON/OFF toggle (green when ON), and a RECONNECT DRIVER button for stuck driver recovery (e.g. ToupTek)
- **Job list** drag-and-drop reorderable, with context menu (Edit / Duplicate / Remove)
- Each job has: filter, count, exposure, gain, offset, binning, frame type (Light/Dark/Flat/Bias), transfer format (FITS/NATIVE/XISF), capture format (RAW/RGB), delay, dither flag, target name
- **+ NEW JOB**: full form with all parameters
- **Presets**: save/load JSON sequences (e.g. 'M31 LRGB night 1')
- **START SEQUENCE**: dialog with 3 options - see chapter 4

## 3.5 Guide

Full PHD2 control:

- Cards: RMS Total, SNR, RMS RA, RMS Dec
- Live RA/Dec tracking error chart (history of the last ~120 samples)
- Buttons: START / STOP / DITHER / FIND STAR / CALIBRATE / CLEAR CAL / PAUSE
- PHD2 equipment: version, pixel scale, calibrated, settling

## 4. Pre-flight pipeline

When you tap **START SEQUENCE** in Capture a dialog appears with **three execution modes**:

### 4.1 FULL OBSERVATION (recommended)

Runs an orchestrated 10-stage pipeline that reproduces what Ekos Scheduler does:

| #  | Stage           | Description                                           | Skip |
|----|-----------------|-------------------------------------------------------|------|
| 1  | resolve_target  | Resolves the name via SIMBAD/astropy → RA/Dec         | -    |
| 2  | slew            | Mount goto+track to the target                        | -    |
| 3  | tracking        | Waits for the mount to be Ok (max 5 min)              | -    |
| 4  | plate_solve     | solve-field on the last frame with $\pm 5^\circ$ hint | opt  |
| 5  | sync_mount      | Syncs the mount on the solve, re-enables tracking     | auto |
| 6  | autofocus       | Iterative HFR loop with V-curve                       | opt  |
| 7  | guide_calibrate | PHD2 clear+recalibrate, wait for Guiding (4 min)      | opt  |
| 8  | guide_start     | PHD2 start guiding, wait for settle (3 min)           | opt  |
| 9  | capture_load    | Loads the .esq into Ekos via DBus                     | -    |
| 10 | capture_started | Starts the Ekos queue                                 | -    |

Capture only starts after every stage has succeeded. The app shows a **live timeline** with colored stage states (grey=pending, amber=running, green=done, red=failed).

**Note:** Optional stages can be enabled in the Observation screen. For the first night I suggest enabling everything. On later cycles you can disable plate solve and calibrate (the slowest ones) for a faster start.

### 4.2 VIA EKOS (loadSequenceQueue)

Generates an .esq file from the Flutter jobs and loads it into the Ekos Capture queue via DBus, then starts. The sequence appears **inside** the Ekos Capture window on the desktop. Ekos handles dither, autofocus on filter change, FITS naming, meridian flip - all of its native workflow.

### 4.3 DIRECT (via INDI)

The app drives the INDI drivers directly without going through Ekos. Faster but Ekos won't see the sequence in its UI.

**Warning:** In DIRECT mode the app does NOT modify Ekos's setup. The bridge intercepts BLOBs as a parallel INDI client via enableBLOB Also, so files are not saved to disk by the bridge - everything stays in RAM.

## 5. Advanced screens

All available from the side drawer (tap the menu icon).

### 5.1 Live View

Full-screen viewer for the last captured frame, with pinch zoom and metadata (HFR, stars, exposure, filter).

### 5.2 Focus

Focuser control with iterative autofocus:

- Manual movement  $\pm 10/100/1000$  steps in/out
- Absolute position with numeric field
- **Iterative autofocus**: set step size, n steps (odd), exposure  $\rightarrow$  START. The bridge takes N shots at different positions, computes HFR for each, finds the minimum, moves to the best position.
- **Live V-curve plot** with colored points and a dashed vertical line on the best position

### 5.3 Align (Plate Solve)

Plate solving via solve-field astrometry.net:

- Shows the last captured frame and its metadata
- Tap **PLATE SOLVE**  $\rightarrow$  the backend launches `solve-field` with hints from the current mount (5° radius), polls status every 2s
- When done, shows RA/Dec/scale extracted via `astropy.wcs.WCS` from the `.wcs` file
- **SYNC MOUNT** button: syncs the mount on the solve result
- Full solve-field output expandable for debug

### 5.4 Observatory

Dome, dust cap, flat panel, weather control:

- Weather card with all parameters (temp, humidity, wind, sky)
- Dome shutter Open/Close
- Dust cap Park/Unpark
- Flat panel toggle + 0-255 intensity slider

### 5.5 Scheduler

Multi-target nightly planner:

- Sky-state card: twilight phase (day/civil/nautical/astronomical/night), sun/moon altitude, lat/lon (auto-detected from the mount)
- Persistent jobs list with RA/Dec, minimum altitude, time window
- +NEW JOB: form with automatic SIMBAD target resolution
- For each job: tap ✓ to live-check conditions (twilight required, current altitude, weather safe)  $\rightarrow$  dialog with the list of issues



## 5.6 Setup / Profiles

Active INDI driver list with **CONNECT/DISCONNECT** toggle (useful to enable drivers like XAGYL Wheel or Weather Watcher when Ekos loaded but did not connect them).

## 5.7 INDI Panel

Exact clone of the KStars/Ekos INDI Control Panel:

- List of every connected device
- Tap a device → all its properties grouped by Group (Main Control, Options, etc.)
- Interactive Switches (ChipToggle), editable Numbers, editable Texts, read-only Lights with colored state
- Live two-way propagation: change here → Ekos sees it; change in Ekos → it appears here
- CONNECT/DISCONNECT button at the top for each device

## 5.8 Files

Browser for FITS files in ~/Pictures/Ekos/:

- Recent files list with auto-stretched thumbnails
- Tap → full-screen preview with zoom
- Light/Dark/Flat/Bias/All filters
- Multi-select with long-press and batch delete

## 5.9 Logs / Activity Log

Two distinct screens:

- **INDI Logs:** live INDI/Ekos message stream, filterable by module
- **Activity Log:** every HTTP request the app makes to the bridge, with millisecond timestamp, color-coded status code, duration, body. Tap for details + copy. Crucial for debugging.

## 5.10 Analyze

Current session timeline:

- Counters: WS events, total properties, devices
- Last frame info (object, filter, exposure, HFR, stars, size)
- Historical PHD2 RMS chart (RA in amber, Dec in cyan)
- INDI message stream, last 20 lines

## 6. Troubleshooting

| Symptom                         | Likely cause                                  | Fix                                                                       |
|---------------------------------|-----------------------------------------------|---------------------------------------------------------------------------|
| App: 'unreachable' on login     | Tailscale not active on the phone             | Check the Tailscale app shows Connected; ping 100.74.22.40 from the phone |
| App: 'auth_failed'              | Wrong token                                   | cat ~/.config/astroarch-bridge/token on the RPi                           |
| App sees 0 devices              | Ekos profile not started                      | Open Ekos on the desktop and start the INDI profile                       |
| Cooler not working (POWER 0%)   | toupbase driver stuck after fast drop         | Tap RECONNECT DRIVER on Capture                                           |
| Sequence does not start in Ekos | UPLOAD_MODE in CLIENT (driver forced)         | Set to BOTH from app v0.2.2+                                              |
| Plate solve fails               | Astrometry index files missing or wrong scope | Check ~/share/kstars/astrometry/, scale hint                              |
| WS frames does not refresh      | WS state disconnected (red banner)            | Manual refresh (Dashboard icon) or reconnect                              |
| FITS files not saved            | UPLOAD_DIR misspelt or directory missing      | App creates /Pictures/Ekos/AstroarchInterface/                            |
| Multi-camera 409 error          | Bridge does not know which camera to use      | Use the Camera dropdown in Capture (default = primary)                    |

### 6.1 Diagnostic commands on the Raspberry Pi

```
# Bridge service status
systemctl --user status astroarch-bridge

# Live log
journalctl --user -u astroarch-bridge -f

# Endpoint test
curl http://localhost:8765/healthz

# Unstuck a frozen INDI driver
indi_setprop 'CAMERA_NAME.CONNECTION.DISCONNECT=On'
sleep 2
indi_setprop 'CAMERA_NAME.CONNECTION.CONNECT=On'

# Inspect cooler state
indi_getprop 'CAMERA_NAME.CCD_TEMPERATURE.CCD_TEMPERATURE_VALUE'
indi_getprop 'CAMERA_NAME.CCD_COOLER_POWER.COOLER_POWER'
```

### 6.2 Diagnostics inside the app

On the Login screen there is a **TEST** button that opens the Diagnostics screen with 7 sequential steps:

- Host resolution (DNS / Tailscale)
- HTTP GET /healthz
- HTTP GET /api/system/info (auth Bearer)
- HTTP GET /api/system/snapshot (payload check)
- WebSocket /ws/state (open)
- WebSocket - first message within 5s
- WebSocket - chunked property\_def reception

Each step shows the duration in milliseconds and the error details on failure. It solves 95% of issues at first try.

## 7. Appendix - REST API

All requests need the header Authorization: Bearer <token> (except /healthz). Responses are always JSON.

### 7.1 System

| Endpoint                 | Method | Description                                          |
|--------------------------|--------|------------------------------------------------------|
| /healthz                 | GET    | Health (no auth)                                     |
| /api/system/info         | GET    | Bridge info (version, author)                        |
| /api/system/snapshot     | GET    | Global state (devices, properties, phd2, last_frame) |
| /api/system/connections  | GET    | INDI/PHD2 connection state                           |
| /api/system/devices      | GET    | INDI device list                                     |
| /api/system/camera_roles | GET    | Identifies primary vs guide camera via PHD2          |
| /api/system/simbad       | GET    | ?name=M31 -> RA/Dec via SIMBAD                       |

### 7.2 Mount, Camera, Focuser, Filter, Guide

| Endpoint                                                                                                                  | Method   | Description                   |
|---------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------|
| /api/mount/status, /goto, /park, /track, /slew, /slew_rate, /abort                                                        | GET/POST | Telescope control             |
| /api/camera/status, /expose, /abort, /cooler, /gain, /offset, /set_type, /transfer_format, /capture_format, /upload_setup | GET/POST | Camera control                |
| /api/focuser/status, /abs, /rel, /abort, /autofocus, /autofocus_iter                                                      | GET/POST | Focuser + iterative autofocus |
| /api/filter_wheel/status, /select                                                                                         | GET/POST | Filter wheel                  |
| /api/guide/status, /start, /stop, /dither, /loop, /clear_calibration, /pass, /dither, /calibrate, /profile                | GET/POST | Guide                         |

### 7.3 Align, Capture/Ekos, Observation, Scheduler, Setup

| Endpoint                                                                                                        | Description                           |
|-----------------------------------------------------------------------------------------------------------------|---------------------------------------|
| /api/align/status, /solve, /solve/{id}/sync_mount, /capture_and_solve, /align, /align_status, /ekos_full_status | Align, solve, capture and solve, Ekos |
| /api/capture/ekos_alive, /ekos_run, /ekos_status, /ekos_abort, /ekos_clear                                      | Ekos via Ekos DBus                    |
| /api/observation/run, /{id}, /{id}/abort                                                                        | Pre-flight pipeline 10 stages         |
| /api/scheduler/sky_state, /jobs, /jobs/{id}/check_conditions, /weather_info                                     | Scheduler                             |
| /api/setup/profiles, /active_drivers                                                                            | Ekos profiles + active drivers        |
| /api/observatory/status, /dome/shutter, /dust_cap, /flat_panel                                                  | Dome + flat panel                     |
| /api/files/recent, /preview, /download, /file (DELETE), /delete_many, /file_browser                             | File browser                          |

### 7.4 INDI panel (control panel clone)

| Endpoint              | Description |
|-----------------------|-------------|
| GET /api/indi/devices | Device list |

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| GET /api/indi/devices/{dev}/properties            | All device properties           |
| GET /api/indi/devices/{dev}/properties/{name}     | Single property                 |
| POST /api/indi/devices/{dev}/properties/{name}    | Set values (Switch/Number/Text) |
| POST /api/indi/devices/{dev}/connect, /disconnect | Connect/disconnect driver       |
| POST /api/indi/refresh                            | Force getProperties             |

## 7.5 WebSocket

| URL                      | What it pushes                                                                               |
|--------------------------|----------------------------------------------------------------------------------------------|
| GET /ws/state?token=...  | snapshot_begin -> N property_def -> snapshot_end (init), then property_def, property_set, pr |
| GET /ws/frames?token=... | For each frame: JSON header {type:frame_meta, size, hfr, ...} followed by JPEG bytes         |

## 8. Changelog

| Version  | Highlights                                                           |
|----------|----------------------------------------------------------------------|
| 0.1.0    | First release: 13 base screens, live WebSocket clone, Tailscale auth |
| 0.1.4    | Chunked WebSocket snapshot (fix 200KB Android payload)               |
| 0.1.5-6  | HTTP/WS cleartext fix + step-by-step diagnostics                     |
| 0.1.7    | Camera primary/guide auto-detection via PHD2                         |
| 0.1.8-10 | Cooler robust toggle + capture sequence + Activity Log               |
| 0.2.0    | Full Ekos clone: 8 modules + multi-job + SIMBAD + autofocus V-curve  |
| 0.2.1    | Plate solving via solve-field + Time scheduler + sky_state           |
| 0.2.3    | Capture via Ekos DBus (loadSequenceQueue + start)                    |
| 0.2.4    | FULL OBSERVATION pipeline with 10 pre-flight stages                  |
| 0.2.7    | Plate solve clone Ekos (full status + bin/gain configurable)         |
| 0.2.10   | Plate solve UI redesign (live preview, big action button)            |
| 0.2.11   | BLOB intercept zero-invasive (parallel INDI client)                  |
| 0.2.12   | Auto-stretch identical to KStars/Ekos (ZScale + asinh MTF)           |

### Future roadmap

- Mosaic planner
- Push notifications (frame done, sequence finished, weather alert)
- Possible upstream contribution to devDucks/astroarch

- Clear skies -

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